

Achyut Niroula

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EXPERIENCE

Apatheia

Jun. 2025 – present

Co-Founder, Founding Software Engineer

North Bay, ON

- Built distributed REST APIs on AWS (EC2, S3, SQS) connecting 8 neural models in an async computer vision pipeline; owned service architecture and deployment for high availability.
- Containerized all microservices on Linux/EC2 with Docker; wrote 9 automated test suites covering frame-level correctness and pipeline edge cases.
- Implemented observability across all model endpoints; authored root-cause reports and API documentation consumed by internal engineering stakeholders.

Nipissing University

May 2023 – Aug. 2025

Web Developer, Research Assistant

North Bay, ON

- Built a database-backed faculty CMS end-to-end using Python, Flask, and SQLite, serving ~70 active users; supported course material distribution, assignment resources, and student discussion and homework submission; doubled as the faculty member's portfolio site.
- Redesigned the frontend with HTML, CSS, and JavaScript; improved Lighthouse performance score to 94+ on a Faculty site serving static content with no client-side rendering overhead.
- Gathered requirements from faculty stakeholders; authored technical documentation for developer and non-technical audiences across the full SDLC.

PROJECTS

JobHunt AI + ResOptimum | Python, FastAPI, React, TypeScript, Docker, Nginx, SQLite, Claude API, pdfplumber

- Built a distributed scraping platform pulling from 7 job boards concurrently with semantic matching and ATS classification; processed 1,200+ listings per run with sub-3-second board response handling via async I/O and connection pooling.
- Engineered ResOptimum as an integrated module: structured LLM prompts with JSON schema validation, role detection, and prompt chaining; REST API ingesting PDF and job description inputs to produce ATS-targeted LaTeX output with automatic one-page fit-check via a second LLM pass.
- Wrote pytest integration tests covering all API endpoints; containerized the full-stack deployment with Docker Compose end-to-end.

LEON (LearnOne) | Python, FastAPI, PostgreSQL, Redis, React, TypeScript

- Built a RAG pipeline embedding codebases into PostgreSQL (pgvector) for real-time Q&A on a FastAPI backend; implemented Redis-backed multi-provider LLM fallback chain for high availability under rate limits.
- Designed a real-time voice pipeline with native audio streaming, VAD-based interruption, and automatic session resumption sustaining conversations beyond connection-length limits.

EDUCATION

Nipissing University

May 2026

B.Sc. Honours Specialization in Computer Science — GPA: 3.7/4.0

North Bay, ON

- Awards: President's Scholarship (2021), Carl Sanders Scholarship (2022, 2023, 2024, 2025)
- Relevant coursework: Distributed Systems (96%), Databases & Data Management (96%), Software Engineering (89%), Artificial Neural Networks (83%)

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

Frameworks & APIs: FastAPI, React, Flask, Node.js, RESTful APIs, microservices, Nginx

Cloud & DevOps: AWS (EC2, S3, SQS, DynamoDB), Google Cloud Run, Docker, Docker Compose, Git, Linux

Databases: PostgreSQL, Redis, SQLite, Firebase (NoSQL)

AI & ML: Computer vision pipelines, RAG, pgvector, LLM integration, semantic similarity

Testing: pytest, JUnit, Playwright, integration and end-to-end testing

Other: System architecture, stakeholder communication, technical writing, agile/SDLC